

12 COMPANY PROFILES

12.1 KEY PLAYERS

12.1.1 CHARLES RIVER LABORATORIES

12.1.1.1 Business overview

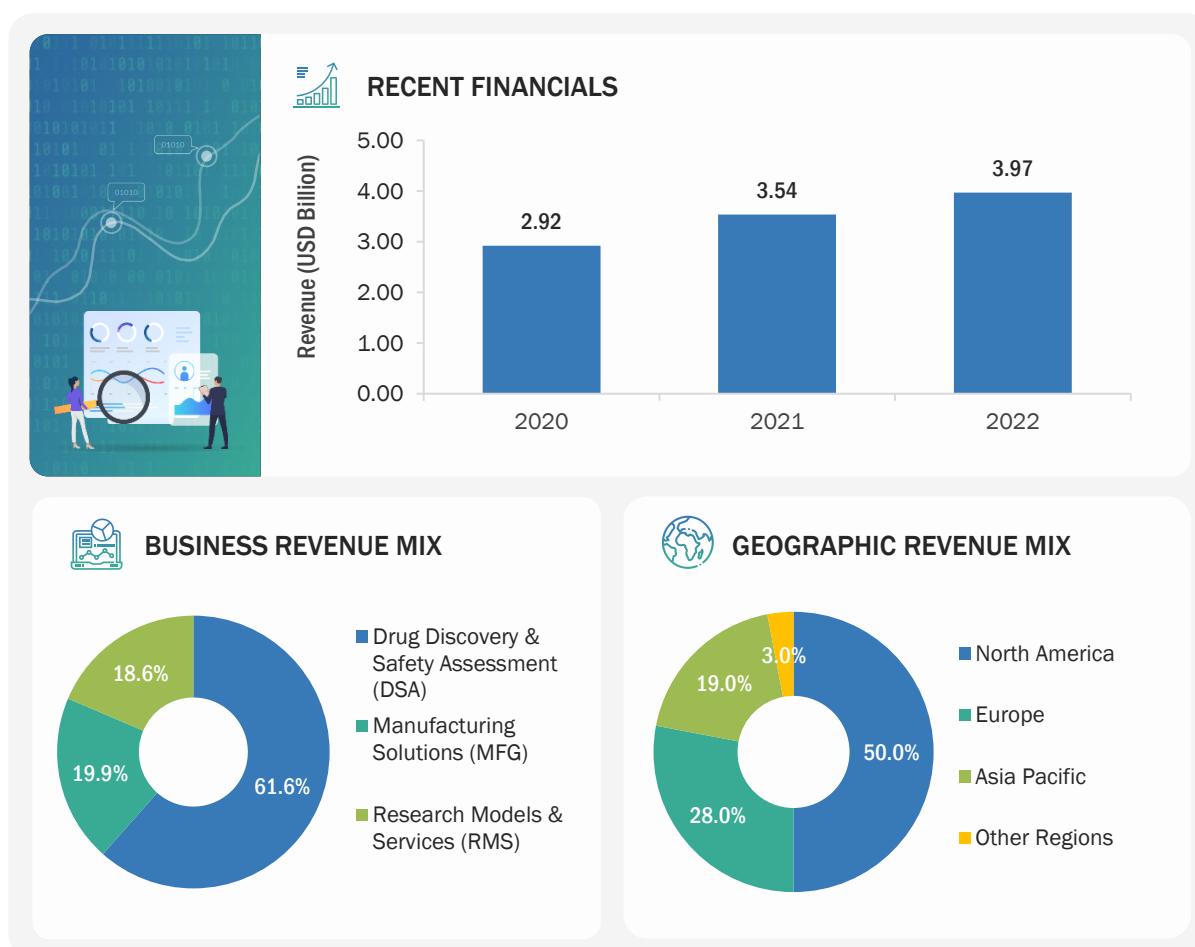
Charles River Laboratories (CRL) is a service and early-stage CRO. The company provides a diverse portfolio of discovery and safety assessment services and offers various products and services to support its clients' manufacturing activities. The company operates through three reporting segments, namely, Research Models and Services (RMS), Drug Discovery and Safety Assessment (DSA), and Manufacturing Support. The company offers microbiome research services (including sequencing services) through all three business segments. CRL offers services to analyze and monitor the health of research models through the RMS segment. The company offers microbiome sequencing services for animal diagnostics.

CRL has 100+ facilities in 20+ countries worldwide. It sells products and services principally through its direct sales and business development teams based in North America, Europe, and Asia. Some of the notable subsidiaries of the company are Cognate Bioservices Inc., Vigene Biosciences Inc., MPI Research, Charles River Laboratories Italia Srl, Charles River Laboratories Japan, and Charles River UK Limited.

TABLE 172 CHARLES RIVER LABORATORIES: BUSINESS OVERVIEW

CHARLES RIVER LABORATORIES	
Year Founded	1947
Country	US
City/State	Massachusetts
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 32 CHARLES RIVER LABORATORIES: COMPANY SNAPSHOT (2022)

Note: The company offers microbiome sequencing services through all three business segments (DSA, RMS, and MFG). Hence, the recent financials, business revenue mix, and geographic revenue mix have been provided for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.1.1.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
NGS Microbiome Sequencing	16S Next Generation Sequencing	The company offers 16S NGS services for microbiome library preparation and sequencing through three service types, namely, Microbiome Basic, Microbiome Plus, and Advanced Analysis.	Animal health monitoring, germ-free animal contamination testing, and human disease modeling.
Microbial Sequencing ID Services (Strain Typing/ Identification Services)	Whole-genome Sequencing (AccuGENX-ID Sanger Sequencing)	- The company offers microbial identification services through its proprietary product line of microbiome products: - Fungal Identification-AccuGENX-ID FunITS - Bacterial Identification-AccuGENX-ID BacSeq - Protein-Coding Gene Sequencing-AccuGENX-ID ProSeq - AccuBLAST Sequencing Data Analysis	Fungal and bacterial identification.

12.1.1.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
October 2020	Partnership	CRL (US)	Fios Genomics Ltd. (UK)	Charles River Laboratories entered into an exclusive partnership with Fios Genomics, a leading provider of bioinformatics data analysis services. Through this partnership, Charles River's clients will have access to Fios Genomics' expertise in bioinformatics, statistics, and biology, to assist in the sourcing and analysis of the high-dimensional, multi-variant datasets associated with drug development, including microarrays, next-generation sequencing (NGS), proteomics, metabolomics, and epigenetics as well as the associated meta-data.	NA

Source: Press Releases

12.1.1.4 MnM view

12.1.1.4.1 Key strengths

CRL is a leading player in the microbiome sequencing services market. Its key strength lies in its strong presence in the adjacent portfolios of research and animal diagnostic services and drug discovery services. The company has been focusing on organic and inorganic growth through expansions and partnerships, as observed in the last three financial years. In October 2020, CRL partnered with Fios Genomics Ltd. (UK) to gain access to its bioinformatics solutions, helping CRL to develop a full spectrum of microbiome profiling/analysis services. In addition, the strong reach of CRL's microbial detection product, Accugenix, has ensured its dominant position in the market.

12.1.1.4.2 Strategic choices

The company focuses on strategic inorganic growth strategies to co-develop advanced technologies and strengthen its research and commercialization capabilities through acquisitions in the microbiome sequencing services market. In August 2012, CRL acquired Accugenix, Inc., a cGMP-compliant contract microbial identification testing provider. This acquisition strengthened CRL's capabilities around state-of-the-art microbial detection services for manufacturing in the biopharmaceutical medical device, nutraceutical, and consumer care industries. In 2023, the company entered into an acquisition agreement with SAMDI Tech for drug discovery research. The acquisition marks the culmination of a partnership between the companies that began in 2018. With its expanding service portfolio, product innovation, and wide geographic presence, the company is expected to grow at a high rate in the microbiome sequencing market during the forecast period.

12.1.1.4.3 Weaknesses and competitive threats

The company faces increased competition from private companies that have dedicated portfolios for microbiome sequencing services across the globe. This may hamper its profitability. The company aims to retain existing customers and obtain new customers and business partners to sustain the competition.

12.1.2 EUROFINS SCIENTIFIC

12.1.2.1 Business overview

Eurofins Scientific is a life science company that provides a range of analytical testing services to its clients across multiple industries. The company operates through only one business segment—Analytical Testing. Different analytical testing services offered under this business segment include Biopharmaceutical Services, Food and Feed Testing, Environment Testing, Clinical Diagnostics, Consumer Product Testing (CPT), and IVD Solutions. The company offers next-generation sequencing services (including microbiome sequencing) through the Biopharmaceutical Services business segment. Eurofins Genomics is part of the Eurofins Scientific Group, which provides NGS, genotyping, gene expression, and custom DNA sequencing, along with oligonucleotides, siRNA, and gene synthesis services.

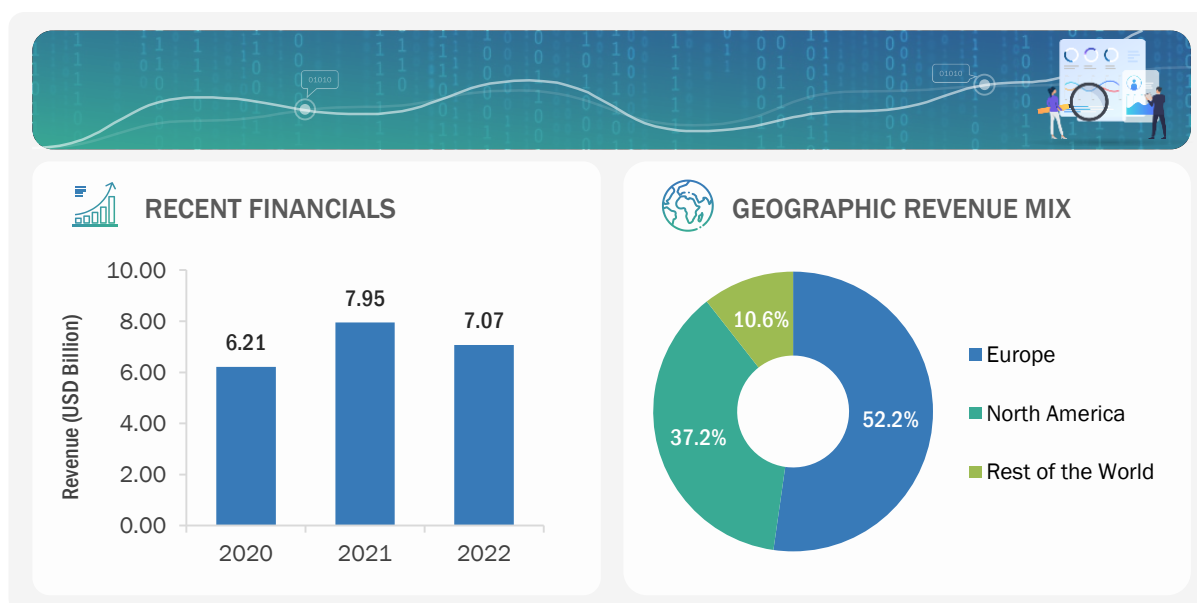
The company has R&D and production sites in the US, Germany, India, and Japan and provides its services to around 200 laboratories across 36 countries in Europe, North America, the Asia Pacific, and Latin America. Some of the company's major subsidiaries are Eurofins BioPharma Product Testing (Shanghai) Co., Ltd. (China), Eurofins Genomics Europe Shared Services GmbH (Germany), Transplant Genomics, Inc. (US), and GATC Biotech SAS (France).

TABLE 173 EUROFINS SCIENTIFIC: BUSINESS OVERVIEW

EUROFINS SCIENTIFIC	
Founded	1987
Country	Luxembourg
City/State	Luxembourg City
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 33 EUROFINS SCIENTIFIC: COMPANY SNAPSHOT (2022)



Note 1: The Business revenue Mix of the company is not available in the public domain. Hence, it is not provided in the company snapshot.

Note 2: The Rest of the World includes countries across South America, Asia, the Middle East, and the Pacific.

Source: Company Website and Annual Reports

12.1.2.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
NGS Microbiome Sequencing/ Profiling	16S Next Generation Sequencing	Eurofins offers microbiome profiling, including sequencing of samples for the below target list: ▪ 18S ▪ plant ITS1 ▪ plant trnL ▪ Different primer pairs for ITS2 ▪ 16S bacteria V4 ▪ 16S bacteria V5-V6 Different primer pairs for: ▪ 16S bacteria V3-V4	Bacterial taxonomic profiling, fungal taxonomic profiling, and archaeal taxonomic profiling.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.2.3 MnM view

12.1.2.3.1 Key strengths

Eurofins Scientific is a prominent player in the market. The company's strong focus on a broad range of customers, technology-enabled solutions, investments in joint ventures and partnerships, and comprehensive quality systems and processes contribute to its market growth. The company provides drug discovery and development services to more than 220,000 customers, including managed care organizations (MCOs), biopharmaceutical companies, governmental agencies, physicians, hospitals and health systems, food & nutritional companies, and independent clinical laboratories.

12.1.2.3.2 Strategic choices

The company focuses on organic and inorganic growth strategies such as acquisitions, expansions, and product innovation. The company started integrating Illumina's NGS platforms in 2019 and has since focused on enhancing its service portfolio around microbiome sequencing. Through this integration, the company expanded its service offerings for microbiome or bacterial analysis, which can be used for the monitoring or assessment of skin microbiomes in order to identify diseases.

12.1.2.3.3 Weaknesses and competitive threats

The company faces increased price competition and competes with various specialized microbiome sequencing service providers across the globe. This may affect its profitability or attractiveness. To sustain its market presence, the company must aim to retain existing customers and obtain new customers and business partners.

12.1.3 BGI GROUP

12.1.3.1 Business overview

Founded in 1999 and headquartered in Shenzhen, China, BGI offers products and services for transcriptomics, genomics, epigenomics, proteomics, meta-omics, single-cell sequencing, bioinformatics analysis, and other services. In the field of genomics, the company offers solutions for exome sequencing, whole-genome sequencing, single-cell DNA sequencing, target region sequencing, de novo sequencing, genotyping, and Sanger sequencing. It also offers sequencing and sample preparation systems and kits through its subsidiary, MGI Tech Co., Ltd. (China).

BGI provides its services and solutions in more than 100 countries across the globe, with a customer base of more than 3,000 institutes across Europe, North America, and the Asia Pacific. BGI operates through nine subsidiaries to promote applications in diverse fields. The company's subsidiaries include Complete Genomics, Inc. (US), BGI Diagnosis (Denmark), and BGI Research (China).

TABLE 174 BGI GROUP: BUSINESS OVERVIEW

BGI GROUP	
Year Founded	1999
Country	China
City/State	Shenzhen
Ownership	Public

Note: BGI Group is listed on the Shenzhen Stock Exchange; however, no financial information is available in the public domain. Hence, financial details are not included in this section.

Source: Company Website

12.1.3.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Metagenomic Sequencing	Whole-genome Shotgun Sequencing	The company provides whole-genome shotgun sequencing for DNA samples, which are provided for complex environmental and human samples.	- Species composition and abundance analysis - Analysis of functional genes - Gene differences between samples - Metabolic pathways

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.4 COSMOSID

12.1.4.1 Business overview

CosmosID is a CLIA-certified and GCP-compliant laboratory that offers microbiome sequencing and bioinformatics solutions suitable for research and clinical applications. The company has a developed NGS facility and designs software solutions that enhance the interpretation of metagenomic data. These solutions are designed to provide strain-level resolution to research professionals. The company offers various microbiome sequencing services, including amplicon sequencing, shallow shotgun sequencing, deep shotgun metagenomic sequencing, and metatranscriptomics sequencing.

CosmosID is a private company with operations in the US. It generates its revenue through one reporting segment, Microbiome Solutions.

TABLE 175 COSMOSID: BUSINESS OVERVIEW

COSMOSID	
Year Founded	2008
Country	US
City/State	Germantown, Maryland
Ownership	Private

Note: CosmosID is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.4.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Amplicon Sequencing	Bacteria/Archaea - 16S rDNA Sequencing	This sequencing service is designed to support amplicon-based metagenomics projects. The service is suitable for research and commercial applications.	Routine testing, large cohort studies, biobanking, population-based studies, and microbiome-based diagnostics.
Shotgun Sequencing	Shallow Shotgun Sequencing	CosmosID offers shallow shotgun sequencing at 3M reads, which yields consistent species and strain-level resolution of bacteria.	Large cohort studies based on the gut microbiome.
	Deep Shotgun Metagenomic Sequencing	In addition to the sequencing services, the company offers validated DNA extraction protocols suitable for shotgun sequencing.	Clinical metagenomics, infectious disease diagnostics, and biomarker discovery.
Metatranscriptomics Sequencing	NGS Library Preparation/ Sequencing	CosmosID makes use of optimized RNA workflows to offer metatranscriptomic services for microbial isolates and other complex samples.	Nutritional studies, drug discovery and development, and probiotics/live biotherapeutic product (LBP) discovery.

Bacterial Isolate Sequencing	Whole-genome Sequencing (WGS)	The company offers end-to-end services for whole-genome sequencing (WGS) of microbial isolates. The sequencing is carried out in CLIA-certified facilities constructed by the company.	Probiotics and LBP characterization, academic research, and clinical microbiology.
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12.1.4.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	SERVICE	DESCRIPTION
January 2023	Metagenomics/ Metabolomics Services	CosmosID Metabolomics Service	The company launched its metabolomics services designed to offer support for multi-omics microbiome research. The service consists of a combination of Liquid Chromatography–Mass Spectrometry (LC-MS) and Gas Chromatography–Mass Spectroscopy (GC-MS).
September 2021	Software & Service Platform	CosmosID-HUB Software Platform	The company announced the addition of a functional analysis module to its online user interface, namely, CosmosID-HUB: Microbiome. This software is combined with microbiome sequencing services offered by CosmosID.

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2023	Partnership	CosmosID (US)	Locus Biosciences (US)	Locus Biosciences announced a long-term partnership with CosmosID to gain the support of the latter for clinical trial studies. CosmosID's support for microbiome analysis is expected to assist Locus Biosciences in strengthening its precision therapeutics platform.	NA
October 2022	Partnership	CosmosID (US)	Microbiome Labs (US)	Microbiome Labs partnered with CosmosID to co-develop gut microbiome tests. This is a stool test designed for microbiome R&D and infectious disease applications.	NA
June 2021	Partnership	CosmosID (US)	Sandwalk BioVentures (Spain)	Sandwalk BioVentures partnered with CosmosID to launch a joint service for Generally-recognized as Safe (GRAS) regulatory genomic reporting. This service combines end-to-end microbial genomics infrastructure and regulatory infrastructure, targeted toward the US probiotics market.	NA

Source: Press Releases

12.1.5 MICROBA

12.1.5.1 Business overview

Microba is engaged in the development of precision microbiome solutions specifically for human gut health. The company has developed an end-to-end analytics platform to measure the human gut microbiome, targeted toward enhancing drug discovery and development. Microba also delivers gut microbiome testing services to clinicians and research professionals.

The company has key operations in Australia and provides microbiome solutions throughout the Asia Pacific region. Microba has three business divisions, namely, Services, Databank, and Therapeutics. The company delivers microbiome sequencing services through the Testing Services segment, which is a part of the Services business division.

TABLE 176 MICROBA: BUSINESS OVERVIEW

MICROBA	
Year Founded	2017
Country	Australia
City/State	Brisbane City, Queensland
Ownership	Public

Note: Financial information for all three years (2020, 2021, and 2022) is not available for Microba in the public domain. Hence, financial details are not included in this section.

Source: Company Website

12.1.5.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Shotgun Sequencing	High-throughput shotgun metagenomic sequencing	Microba uses the Illumina NovaSeq 6000 system to deliver reproducible, high-quality metagenomic sequence data via shotgun metagenomic sequencing.	High-resolution taxonomic and functional data profiling.

12.1.5.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	SERVICE	DESCRIPTION
March 2023	Human Gut Microbiome Testing	Metagenomic Sequencing and Analysis	Microba launched a new human gut microbiome testing service for comprehensive gastrointestinal testing solutions. The solutions are accessible to healthcare professionals in Australia through a different brand name, Co-Biome.

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
June 2021	Partnership	Microba (Australia)	Illumina, Inc. (US)	Microba announced a partnership with Illumina to advance research activities in human gut microbiome health and disease. The partnership is expected to combine NGS tools provided by Illumina with Microba's proprietary analysis platform to develop accurate metagenomic databases.	NA

Source: Press Releases

12.1.6 QIAGEN N.V.

12.1.6.1 Business overview

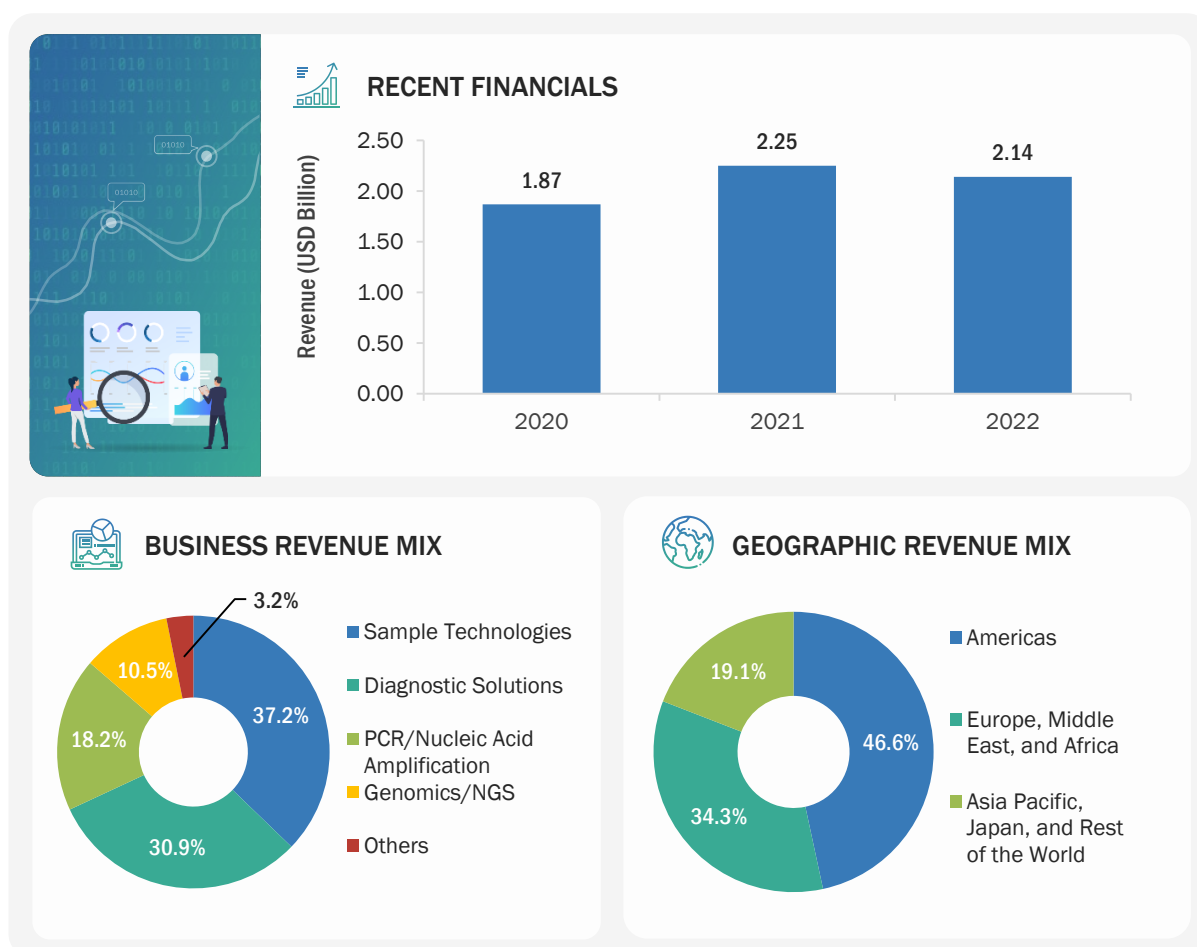
QIAGEN is a leading global provider of sample-to-insight molecular diagnostics and sample preparation technologies. The company operates through a single business segment and generates its revenue through different product categories, namely, Sample Technologies, Diagnostic Solutions, PCR/Nucleic Acid Amplification, Genomics/NGS, and Others. The company offers a wide range of instruments and assay kits for genetic testing and genomic services through all segments and end-to-end microbiome profiling (including sequencing) services through the Genomics segment.

The company operates across Europe, the Americas, the Middle East, Africa, and Asia. Some of the company's major subsidiaries are Amnisure International, LLC (US), Cellestis Pty. Ltd. (Australia), QIAGEN Deutschland Holding GmbH (Germany), QIAGEN Marseille S.A.S. (France), QIAGEN K.K. (Japan), and QIAGEN China (Shanghai) Co. Ltd. (China).

TABLE 177 QIAGEN N.V.: BUSINESS OVERVIEW

QIAGEN N.V.	
Founded	1984
Country	Germany
City/State	Hilden
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 34 QIAGEN N.V.: COMPANY SNAPSHOT (2022)

Note: The company provides microbiome sequencing services through the Genomics/NGS business segment. The information for the Recent Financials, Business Revenue Mix, and Geographic Revenue Mix was not available for the Genomics/NGS. Hence, these sections have been provided for the overall company.

Source: Company Website and Annual Reports

12.1.6.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
NGS Microbiome Sequencing	16S/ITS Microbiome Sequencing	QIAGEN offers a sample-to-insight profile panel, which includes services around sample disruption, sample preparation, NGS library preparation, sequencing, data analysis, and interpretation.	Bacterial and fungal profiling.

12.1.6.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2022	Partnership	QIAGEN (Germany)	Element Biosciences (US)	QIAGEN announced a partnership with Element Biosciences to validate its leading NGS library prep platform and AVITI System-associated offerings. This partnership strengthened AVITI System capabilities through QIAseq target enrichment panels. QIAseq products aid in the detection and characterization of changes in the genome, transcriptome, methylome, and microbiome.	NA

Source: Press Releases

12.1.7 MICROBIOME INSIGHTS

12.1.7.1 Business overview

Microbiome Insights was established through the partnership of the Life Science Institute and the Personalized Medicine Initiative. The company offers end-to-end services for comprehensive microbiome analysis, including different services for microbiome sequencing. Microbiome Insights operates in a single business unit, focused entirely on microbiome analysis/profiling services. These services are offered for different applications, including cosmetics & dermatology, food & beverage, pharma & human health, and agriculture & animal health.

TABLE 178 MICROBIOME INSIGHTS: BUSINESS OVERVIEW

MICROBIOME INSIGHTS	
Year Founded	2015
Country	Canada
City/State	Richmond
Ownership	Private

Note: Microbiome Insights is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.7.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Gene Sequencing	16S & ITS2 rRNA Sequencing Service	Microbiome Insights offers 16S & ITS2 rRNA sequencing service with two primer options—16S V4 and 16S V1-V3.	Bacterial taxonomy analysis and fungal taxonomy analysis.
Shotgun Sequencing	Shotgun Metagenomic Sequencing	The company uses sequencing instruments manufactured by Illumina Inc., namely MiSeq, NextSeq, and HiSeq, to offer microbiome shotgun metagenomic sequencing services.	Taxonomic profiling and functional analysis.
Shotgun Sequencing	Shallow Shotgun Metagenome Sequencing (SSMS)	This service combines different samples in a single sequencing run via a modified protocol. This protocol uses a lower volume of reagents for sequencing. library preparation	Taxonomic profiling and functional analysis.

12.1.7.3 Recent developments

OTHER DEVELOPMENTS

MONTH & YEAR	DEVELOPMENT TYPE	COUNTRY/REGION	DESCRIPTION
November 2021	Expansion	Ireland	Microbiome Insights announced the opening of an office in Dublin, Ireland. This expansion was targeted to serve the company's European clients efficiently. This expansion was also expected to support the increasing number of clients across European countries (UK, Germany, France, Ireland, Portugal, Spain, Austria, and Sweden).

Source: Press Releases

12.1.8 BASECLEAR

12.1.8.1 Business overview

BaseClear develops microbial genomic solutions that enhance the understanding and use of microorganisms. The services offered by the company span from product improvement to regulatory approvals. These services include study design consultancy, sampling and logistics, microbiology, genomics, advanced bioinformatics & biostatistics, microbiome analysis, and regulatory affairs. All of the above services are targeted toward microbiome analysis, including sequencing services. Key product lines of the company related to microbiomes include human health, animal health, skin health, and personal care. BaseClear performs in-house DNA research and analysis, followed by data interpretation services.

BaseClear offers services in Europe, which range from sample collection and study design to analysis.

TABLE 179 BASECLEAR: BUSINESS OVERVIEW

BASECLEAR	
Year Founded	1993
Country	Netherlands
City/State	Leiden
Ownership	Private

Note: BaseClear is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

12.1.8.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Microbial Profiling Service	16S and ITS Sequencing	BaseClear offers complete bioinformatics solutions for microbial profiling analysis. Standard solutions for microbial profiling include 16S and/or ITS amplicon analysis for taxonomic classification and relative frequencies. Illumina MiSeq reads are merged into overlapping pseudo-reads and subsequently aligned against a 16S database.	Bacteria and fungal profiling in samples of food, probiotics, water, microbiome, biofilm, fermentation, and soil.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023

12.1.9 CD GENOMICS

12.1.9.1 Business overview

CD Genomics is a service-based company providing genomics and bioinformatics service solutions through advanced technologies such as next-generation sequencing, Sanger sequencing, PCR and microarray services. The company provides end-to-end microbiome sequencing services, which include sample collection and preparation, DNA extraction, library preparation and sequencing, and bioinformatics analysis, along with providing detailed reports of microbiome sequencing results. The company provides microbiome sequencing services to end users such as academic research institutes, biotechnology companies, pharmaceuticals and biopharmaceuticals, and clinical diagnostics.

The company provides services across North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Some of the major subsidiaries of CD Genomics are CD Genomics Bioinformatics Technology Co., Ltd. (China), CD Genomics Europe Limited (UK), CD Genomics Inc. (US), CD Genomics (Suzhou) Co., Ltd. (China), CD Genomics (Guangzhou) Co., Ltd. (China), CD Genomics (Hong Kong) Limited (Hong Kong).

TABLE 180 CD GENOMICS: COMPANY OVERVIEW

CD GENOMICS	
Founded	2004
Country	US
City/State	New York
Ownership Type	Private

Note: CD Genomics is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.9.2 Services offered

SERVICE TYPE	SERVICE	DESCRIPTION	APPLICATION
Microbial Genomics	16S/18S/ITS Amplicon Sequencing	The company provides DNA extraction, library preparation, sequencing, and bioinformatics analysis for amplicon sequencing.	Microbial identification and phylogeny studies of samples from complicated microbiomes or environments.
Microbial Genomics	Metagenomic Shotgun Sequencing	The company offers DNA extraction, library preparation, sequencing, and bioinformatics analysis for metagenomic shotgun sequencing.	Taxonomic and biological functional characterization of polymicrobial communities.
Microbial Genomics	Viral Metagenomic Sequencing	The company offers DNA extraction, library preparation, sequencing, and bioinformatics analysis for viral metagenomic sequencing.	Analysis of taxonomic composition and structure of viral communities.
Microbial Genomics	Metatranscriptomic Sequencing	The company offers DNA extraction, Library preparation, sequencing and bioinformatics analysis for metatranscriptomic sequencing.	Analysis of active biochemical functions for identifying biomarkers and expression signatures.

Microbial Genomics	Microbial Whole-genome Sequencing	The company offers DNA extraction, Library preparation, sequencing and bioinformatics analysis for microbial whole-genome sequencing.	A comprehensive evaluation of all genetic features of an isolated microorganism.
Microbial Genomics	Absolute Quantitative 16S/18S/ITS Amplicon Sequencing	The company offers DNA extraction, library preparation, sequencing, and bioinformatics analysis for microorganism quantification.	Study of composition and dynamic changes of microbiomes in complex environmental samples.
Microbial Genomics	Microbial Identification	The company offers whole-genome sequencing, Sanger sequencing, and 16S/ITS/18S amplicon sequencing for microbial identification.	Detect, identify, and monitor microbes.
Microbial Identification	Total Microbial Quantification	The company offers qPCR-based microbial quantification services.	PCR quantification is done prior to sequencing a sample of low biomass to determine if the results will have positive amplification.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

Source: Company Website

12.1.10 ZYMO RESEARCH CORPORATION

12.1.10.1 Business overview

Zymo Research is a biotechnology company that provides products and services for molecular biology applications. The major operating segments of the company are DNA and RNA Purification, Epigenetics, Microbiomics, Next-generation Sequencing, Clinical Diagnostics, and Custom Services. Through the Microbiomics segment, the company provides services and products for sample collection and preservation, microbial standards, DNA and RNA extraction kits, library preparation kits, and microbiome sequencing services.

Zymo Research provides microbiome sequencing services to customers worldwide with locations across the US, Germany, China, Japan, the UK, and Australia.

TABLE 181 ZYMO RESEARCH CORPORATION: COMPANY OVERVIEW

ZYMO RESEARCH CORPORATION	
Founded	1994
Country	US
City/State	California
Ownership Type	Private

Note: Zymo Research Corporation is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.10.2 Services offered

SERVICE TYPE	SERVICE	DESCRIPTION	APPLICATION
Microbiome Analysis Services	16S/ITS Amplicon Sequencing Service	The company offers the sequencing of 16S rRNA genes to identify and characterize bacterial and archaeal communities in a sample by providing bioinformatics analysis and raw sequencing data.	Detection and identification of bacteria.
Microbiome Analysis Services	Full-Length 16S Sequencing Service	The company offers full-length sequencing through PacBio HiFi sequencing and bioinformatic analysis.	Bacterial profiling
Microbiome Analysis Services	Shotgun Metagenomic Sequencing Service	The company offers shotgun metagenomic sequencing for the comprehensive profile of microbial communities, including bacteria, archaea, viruses, and fungi, as well as functional information.	Taxonomic identification
Microbiome Analysis Services	Metatranscriptomic Sequencing	The company offers high-throughput sequencing for understanding gene expression patterns of microbial communities in various environments, including soil, water, and human microbiomes.	Analysis, disease diagnosis, biotechnology research, and environmental monitoring.

Microbiome Analysis Services	Custom Solutions for Industry Partners	The company offers customized microbiome sequencing services according to the specific research needs of end users.	Academic research and health & commercial testing.
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Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

Source: Company Website

12.1.11 ORASURE TECHNOLOGIES

12.1.11.1 Business overview

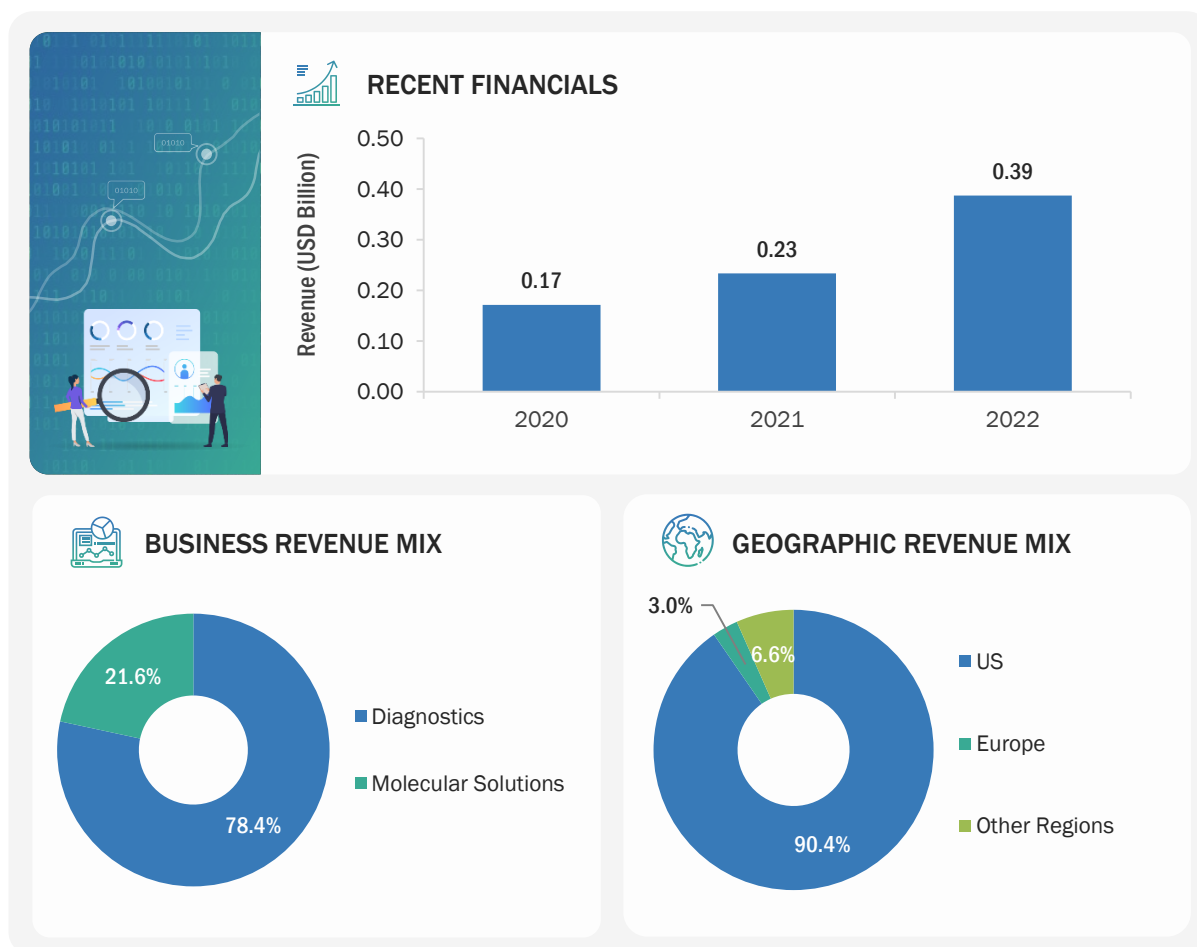
OraSure Technologies is engaged in the manufacturing and distribution of sample collection devices, sample stabilization devices, and rapid diagnostic tests. Key customers of the company include professionals from research institutes, clinical laboratories, hospitals & clinics, physicians' offices, government agencies, pharmaceutical entities, and direct consumers. In February 2023, OraSure Technologies announced a corporate restructuring that combines its innovation teams with commercial expertise to develop multiple product lines. This restructuring is expected to accelerate innovation, enhance customer experience, and result in operational synergies.

The company has locations in the US, Canada, and Europe and operates through its wholly owned subsidiaries, namely DNA Genotek Inc. (Canada), Diversigen (US), and Novosanis (Romania). Through these subsidiaries, OraSure Technologies offers end-to-end diagnostic and research solutions to its customers, including microbiome sequencing services. OraSure Technologies offers microbiome sequencing services via its subsidiary Diversigen (US). Diversigen offers microbiome sequencing and analysis to customers from academics, research, and direct-to-consumer companies. Microbiome profiling services designed by the company serve researchers in pharmaceutical and agriculture R&D, environmental monitoring, cosmetics development, pet nutrition, direct-to-consumer testing, and other industries.

TABLE 182 ORASURE TECHNOLOGIES: BUSINESS OVERVIEW

ORASURE TECHNOLOGIES	
Year Founded	1987
Country	US
City/State	Pennsylvania
Ownership	Public

Source: Company Website

FIGURE 35 ORASURE TECHNOLOGIES: COMPANY SNAPSHOT (2022)

Source: Company Website and Annual Reports

12.1.11.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Microbial Amplicon Sequencing	BenchMark Amplicon Sequencing	BenchMark is a proprietary technology designed for optimum amplicon sequencing to profile/sequence microbiomes. This service also includes a complimentary Core Analysis Report, which provides fully annotated data tables, interactive visualizations, and all the raw data.	This service is suitable for both high and low-biomass samples. It is especially suitable for low biomass samples, including soil, surface swabs, skin, or biopsy samples.
Shotgun Metagenomic Sequencing	BoosterShot Shallow Shotgun Metagenomic Sequencing	The BoosterShot approach is based on shallow shotgun sequencing that offers full species-level resolution and functional profiling attainable for large-scale studies.	This service offers species-level taxonomic profiling and functional profiling of biological pathways in a particular sample.
	DEEPSEQ Deep Shotgun Metagenomic Sequencing	DeepSeq is based on the deep shotgun sequencing approach, which generates big data and uncovers the signatures in microbiome samples.	This service helps in species-level taxonomic profiling and functional profiling of biological pathways along with strain coverage information.

Whole-genome Sequencing	STRAINVIEW Whole-genome Sequencing of Microbial Isolates	High-throughput method for generating draft-level genomes of hundreds or thousands of cultured isolates in parallel.	Strainview enables the characterization of the entire strain collection comprehensively and efficiently.
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Source: Company Website

12.1.11.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	SERVICE	DESCRIPTION
March 2022	Metatranscriptomic Sequencing	Metatranscriptomic Sequencing and Analysis	Diversigen announced the launch of a new service, metatranscriptomic sequencing and analysis of the gut microbiome. This service will assist research professionals in understanding gene expression while gaining insights into both function and mechanism of microbiomes in samples.

Source: Press Releases

12.1.12 MR DNA

12.1.12.1 Business overview

Molecular Research LP (MR DNA) offers next-generation sequencing (NGS) and bioinformatics services through collaborating with instrument/system manufacturers, such as Illumina (US), Applied Biosystems (US), and Pacific Biosciences (US). The company offers a wide array of sequencing services suitable for sequencing small microbial genomes, larger eukaryote genomes, transcriptomes, whole genomes, metagenomes, metatranscriptomes, and exomes. MR DNA is engaged in developing microbiome research solutions such as 16S sequencing services, ITS sequencing, 18S sequencing, Cytochrome c oxidase (COI), or any other type of diversity assays. The company also assists in generating the required data from small or large sequencing projects as a part of an end-to-end service solution.

TABLE 183 MR DNA: BUSINESS OVERVIEW

MR DNA	
Year Founded	2011
Country	US
City/State	Texas
Ownership	Private

Note: MR DNA is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.12.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Amplicon Sequencing Services	16S Ribosomal Sequencing	MR DNA has an extensive in-house assay collection for 16S sequencing, 18S sequencing, and ITS sequencing.	Taxonomic profiling of bacterial, archaeal, and fungal diversity in samples.
Transcriptome Sequencing	PolyA & RNA Sequencing/ Transcriptome Sequencing	MR DNA offers a number of sequencing services for RNA samples.	This is a comprehensive sequencing service that includes RNA extraction and sequencing.
Whole-genome Sequencing	Whole-genome Sequencing (WGS)	MRDNA has access to a number of NGS platforms, including Illumina NovaSeq 6000, Illumina HiSeq, and the PacBio Sequel IIe, through which the company offers WGS.	Taxonomic profiling of bacterial, archaeal, and fungal diversity in samples.
Metagenome Sequencing	Shotgun Metagenome Sequencing	This service is able to target all genes from a number of different samples found in the environment.	Microbial diversity analysis in environmental samples.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.13 EREMID GENOMIC SERVICES

12.1.13.1 Business overview

Eremid Genomic Services is a contract research organization (CRO) engaged in the research and development of products for food & nutrition, agbiotech, and human health. The company's service offerings focus on genomics, leveraging expertise in a range of advanced technologies. Eremid Genomic offers comprehensive services, including scientific consultation, experimental design, sample preparation, sample processing, data analysis, and project reporting.

TABLE 184 EREMID GENOMIC SERVICES: BUSINESS OVERVIEW

EREMID GENOMIC SERVICES	
Year Founded	NA
Country	US
City/State	North Carolina
Ownership	Private

Note: Eremid Genomic Services is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.13.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Microbiome Profiling/Sequencing	Microbiome Species Profiling (16S/18S/ITS)	This portfolio includes 16S, 18S, and internal transcribed spacer (ITS) sequencing services. The	Helps in taxonomic and molecular phylogenetic studies. The characterization of complex microbial strains, genes, and metabolic pathways.
	Microbiome Functional Profiling	shotgun sequencing service offering helps in microbiome functional profiling.	

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.14 CLINICAL-MICROBIOMICS A/S

12.1.14.1 Business overview

Clinical Microbiomics offers microbiome analysis (including sequencing services) for animal health studies, clinical studies, and preclinical studies. The company's portfolio includes DNA extraction, profiling, sequencing profiling, and assessment of clinical data along with the results. The company's proprietary product line for microbiome sequencing includes 16S rDNA amplicon sequencing and shotgun metagenomic sequencing. Clinical Microbiomics has scientific expertise across bioinformatics, microbiology, antibiotic resistance, machine learning, cancer genomics, and preclinical and clinical studies. The company's customers include academic institutes, pharmaceutical companies, nutrition companies, and biotechnology companies developing microbiome-targeted or microbiome-based products. Clinical Microbiomics offers end-to-end microbiome analysis solutions that span from sample preparation and project planning to the final report and data analysis.

TABLE 185 CLINICAL-MICROBIOMICS A/S: BUSINESS OVERVIEW

CLINICAL-MICROBIOMICS A/S	
Year Founded	2015
Country	Denmark
City/State	Copenhagen
Ownership	Private

Note: Clinical Microbiomics is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.14.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Amplicon/16S Microbiome Analysis	16S Microbiome Analysis (Sequencing & data analysis)	The company offers 16S sequencing services using the MiSeq platform (Illumina) coupled with the pipeline of 16S bioinformatics.	- Characterization of bacterial populations in clinical and preclinical research. - Taxonomical analysis for research and discovery. - Bacterial species identification for rapid clinical research and infectious disease research.
Custom Microbiome Analysis	Shotgun Metagenomic Sequencing	This service provides tailored microbiome analysis. Assistance is offered throughout the project, from project design to analysis of the output.	- Microbiome drug discovery - Pre- and probiotics - Microbiome-drug interactions - Biomarker discovery - Strain tracking - Multi-omics integration - De-novo diversity discovery - Responder and non-responder stratification

12.1.14.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
October 2022	Acquisition	Clinical-Microbiomics (Denmark)	MS-Omics ApS (Denmark)	Clinical-Microbiomics announced the acquisition of MS-Omics, a metabolomics service provider. This acquisition will add value to the expertise of Clinical Microbiomics in profiling microbiomes for clinical applications.	NA

Source: Press Releases

OTHER DEVELOPMENTS

MONTH & YEAR	DEVELOPMENT TYPE	DESCRIPTION
October 2022	Funding	Seventure Partners, a microbiome venture capital, announced an investment of USD 11.05 million in Clinical-Microbiomics A/S.

Source: Press Releases

12.1.15 NOVOGENE CO., LTD.

12.1.15.1 Business overview

Novogene is a provider of genomic services and solutions based on NGS and bioinformatics technologies. The company offers a wide range of NGS-based services for human, plant, animal, and microbial DNA and RNA analysis. Novogene has a presence in 70 countries across the globe, with nearly 6,100 customers. Tianjin Novogene Bioinformatics Technology Co., Ltd. (China), Novogene (HK) Company Limited (China), Novogene Corporation Inc. (US), Novogene (UK) Company Limited (UK), Novogene Netherlands BV (Netherlands), and Novogene Japan K.K. (Japan) are the subsidiaries of the company.

TABLE 186 NOVOGENE CO., LTD.: BUSINESS OVERVIEW

NOVOGENE CO., LTD.	
Year Founded	2011
Country	China
City/State	Beijing
Ownership	Private

Note: Novogene Co., Ltd. is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.15.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
De novo Sequencing	Microbial De Novo Sequencing	The company offers these services using PacBio and Illumina platforms. This service is multifaceted and includes sequencing, survey, draft map, complete map, and fine map of genomes.	▪ Virulence research ▪ Molecular markers ▪ Vaccine development ▪ Drug resistance ▪ Microbial evolution ▪ Epidemiology

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.2 OTHER PLAYERS

12.2.1 EZBIOME

EZBIOME	
Year Founded	2020
Headquarters - Country	US
Headquarters – City/State	Maryland
Ownership	Private
Business Overview	The company provides a comprehensive service portfolio that covers sample preparation services and discovery solutions for microbiome research. The service offerings of the company help in microbial identification, microbial genomics, and cross-disciplinary microbiome research. EzBiome has developed a full suite of sample-to-discovery microbiome sequencing services with EzBioCloud bioinformatics for cross-disciplinary research and studies.
Services	Amplicon Sequencing, Shallow Shotgun Sequencing, and Deep Shotgun Sequencing
Geographical Presence	US

12.2.2 BOSTER BIOLOGICAL TECHNOLOGY

BOSTER BIOLOGICAL TECHNOLOGY	
Year Founded	1993
Headquarters - Country	US
Headquarters – City/State	California
Ownership	Private
Business overview	Boster Biological Technology provides an end-to-end service for microbiome analysis. It has developed optimized workflows for accurate microbiome quantification from sample collection to bioinformatics analysis. The company also provides data analysis and interpretation services with a wide range of microbiome reference standards.
Services	▪ 16S/ITS Amplicon Sequencing ▪ 18S Eukaryote Sequencing ▪ Shotgun Metagenomic Sequencing ▪ Metatranscriptomic Sequencing ▪ Customized Microbiome Analysis
Geographical Presence	US